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GENERATIVE DIGITAL TWIN FOR SUPPLY CHAIN DISRUPTION SIMULATION AND RESILIENCE PLANNING

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ABSTRACT

- Supply chain disruptions can occur due to supplier failures, transportation delays, natural disasters, and demand fluctuations.
- Develops a Generative Digital Twin to create a virtual model of the supply chain.
- Uses Generative AI to simulate realistic disruption scenarios.
- Analyzes the impact on inventory, logistics, transportation, and delivery performance.
- Identifies potential risks and evaluates different disruption scenarios.
- Supports proactive planning and faster decision-making.
- Improves supply chain resilience, operational efficiency, and business continuity.

INTRODUCTION

- Supply chains are essential for delivering products but are vulnerable to unexpected disruptions.
- Digital Twin technology creates a virtual model to monitor and simulate supply chain operations.
- Generative AI generates realistic disruption scenarios for effective analysis and planning.
- The system analyzes the impact of disruptions on inventory, logistics, and delivery performance.
- It recommends recovery strategies to improve resilience and business continuity.
- The solution supports better decision-making and reduces operational risks.

LITERATURE SURVEY

Author(s), Year, Journal/Proceedings	Title	Key Findings
P. Bhambri & H. Sharma (2026) , <i>Next-Gen Supply Chains (Wiley)</i>	Generative AI's Impact on Supply Chain Decision-Making	Demonstrated that Generative AI enhances supply chain decision-making through scenario generation, predictive analytics, and digital twin integration.
Latifa Benhamou, Vincent Giard & Samir Lamouri (2026) , <i>International Journal of Production Economics</i>	Digital Twins in Supply Chain Management: Scope and Methodological Issues	Highlighted the role of Digital Twins in real-time monitoring, disruption prediction, and adaptive supply chain management while identifying current implementation challenges.
Dmitry Ivanov (2026) , <i>International Journal of Production Research</i>	Agentic Digital Twins: Bridging Model-Based and AI-Driven Decision-Making Support	Proposed AI-powered digital twins that combine simulation with intelligent decision support for resilient and autonomous supply chain operations.
Tho V. Le & Ruoling Fan (2023) , <i>arXiv</i>	Digital Twins for Logistics and Supply Chain Systems: Literature Review, Conceptual Framework, Research Potential, and Practical Challenges	Presented a comprehensive framework for applying digital twins in logistics, emphasizing improved visibility, resilience, and operational efficiency..
Keqi Wang et al. (2020) , <i>arXiv</i>	Simulation-Based Digital Twin Development for Blockchain Enabled End-to-End Industrial Hemp Supply Chain Risk Management	Developed a simulation-based digital twin integrated with blockchain to improve supply chain transparency, risk management, and decision-making.

RESEARCH GAP

- Existing systems have limited use of **Generative AI** with **Digital Twin** technology.
- Most Digital Twin systems only monitor the supply chain and do not support recovery planning.
- Current methods cannot effectively handle multiple supply chain disruptions at the same time.
- Recovery strategies such as alternative suppliers and routes are not well optimized.
- There is a need for an intelligent system to improve supply chain resilience and decision-making.

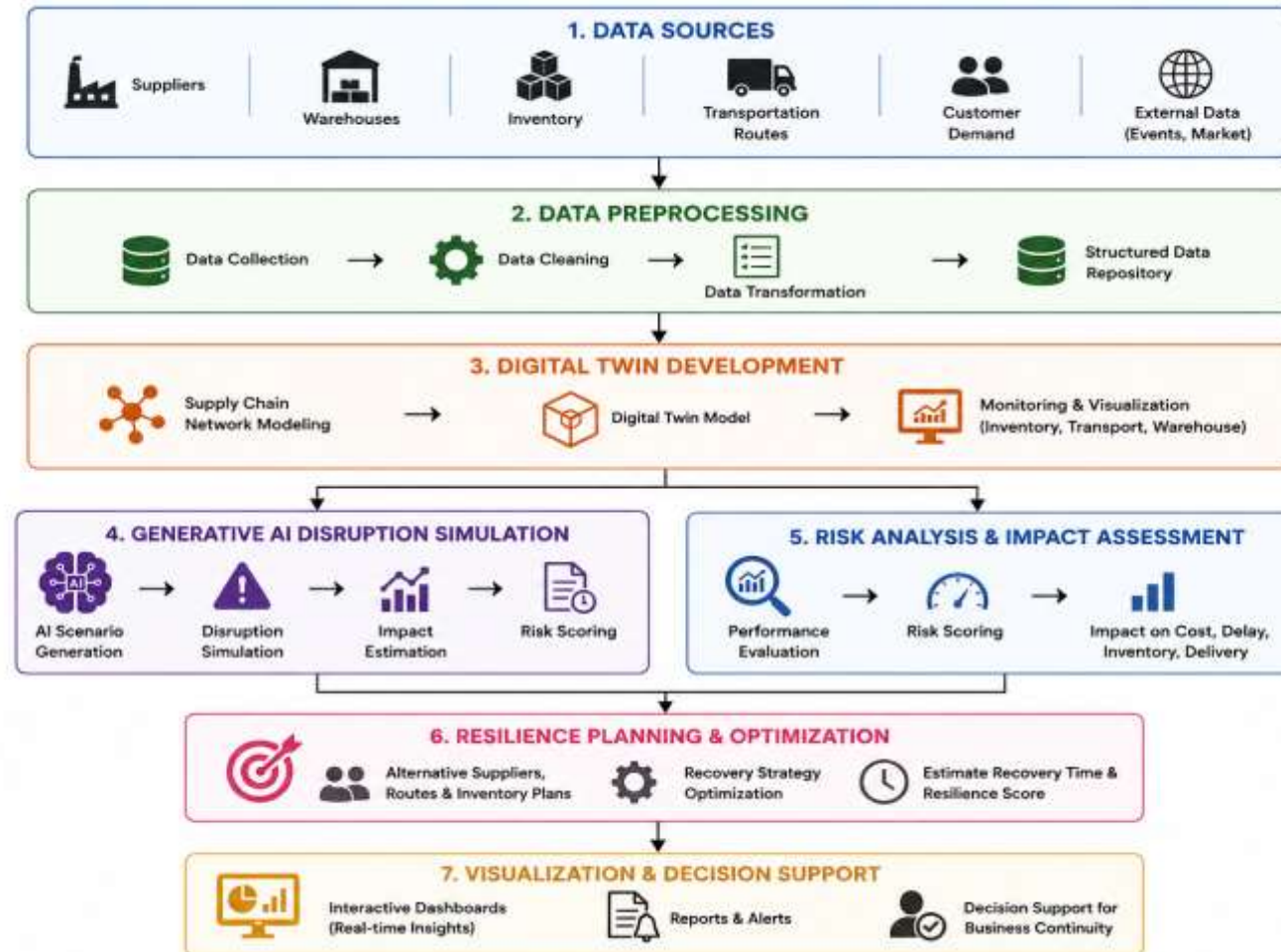
PROBLEM STATEMENT

- Existing Digital Twin systems mainly focus on monitoring and visualization, with limited AI-based disruption prediction.
- Current approaches often simulate only single disruptions instead of multiple real-world disruption scenarios.
- There is a need for an AI-powered framework that can simulate disruptions, analyze their impact, and recommend effective resilience strategies.
- Such a system can improve decision-making, reduce risks, and strengthen supply chain resilience.

OBJECTIVES

- Develop a Digital Twin model of the supply chain.
- Use Generative AI to simulate realistic disruption scenarios.
- Analyze the impact of disruptions on inventory, transportation, cost, and delivery.
- Identify potential risks and assess supply chain performance under different scenarios.
- Improve decision-making and ensure business continuity.
- Enhance overall supply chain resilience and operational efficiency.

SYSTEM ARCHITECTURE



MODULE-1: SUPPLY CHAIN DIGITAL TWIN DEVELOPMENT

- Collect supply chain data, including suppliers, warehouses, inventory, transportation, and customer demand.
- Clean and organize the data for accuracy and consistency.
- Create a **Digital Twin** representing the real-world supply chain.
- Connect all supply chain components to visualize product and information flow.
- Monitor inventory, transportation, and warehouse operations in real time.
- Prepare the Digital Twin for disruption simulation and analysis.

MODULE-2: GENERATIVE AI – BASED DISRUPTION SIMULATION

- Generate realistic supply chain disruption scenarios using **Generative AI**.
- Simulate supplier failures, transportation delays, and demand fluctuations.
- Analyze the impact of disruptions on inventory, delivery time, and operational cost.
- Identify critical risks and assign risk scores to each disruption.
- Evaluate different disruption scenarios for effective risk assessment.
- Transfer simulation results to the resilience planning module.

MODULE-3: RESILIENCE PLANNING AND DECISION SUPPORT

- Analyze disruption results to identify the best recovery strategies.
- Recommend alternative suppliers, transportation routes, and inventory plans.
- Optimize recovery plans to reduce delays and operational costs.
- Estimate recovery time and evaluate overall supply chain resilience.
- Display recommendations through dashboards and reports.
- Support decision-makers in maintaining business continuity during disruptions.

ALGORITHMS/TECHNIQUES USED:

Algorithm/Technique	Purpose
Digital Twin Modeling	Creates a virtual representation of the supply chain.
Large Language Model(LLM)	Generates realistic supply chain disruption scenarios.
Risk Assessment	Evaluates the impact and severity of disruptions.
Dijkstra's Algorithm	Finds the best alternative transportation routes.
Optimization Technique	Recommends optimal recovery and resilience strategies.

TOOLS & TECHNOLOGIES:

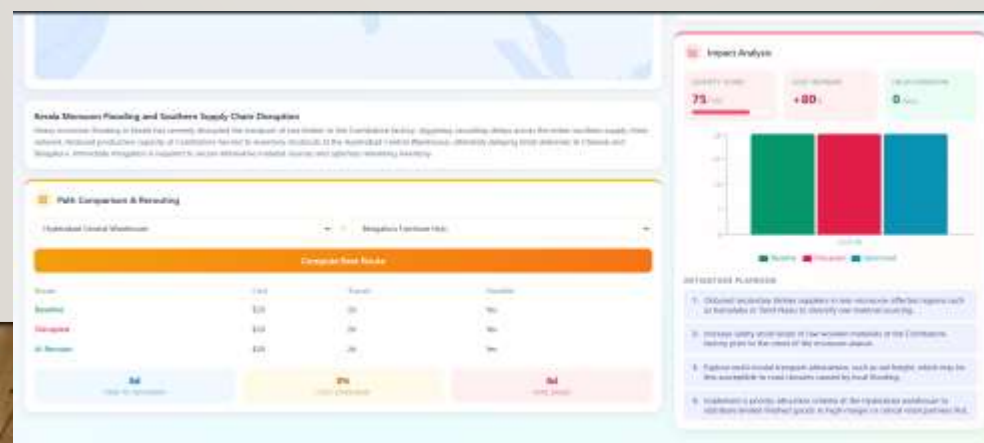
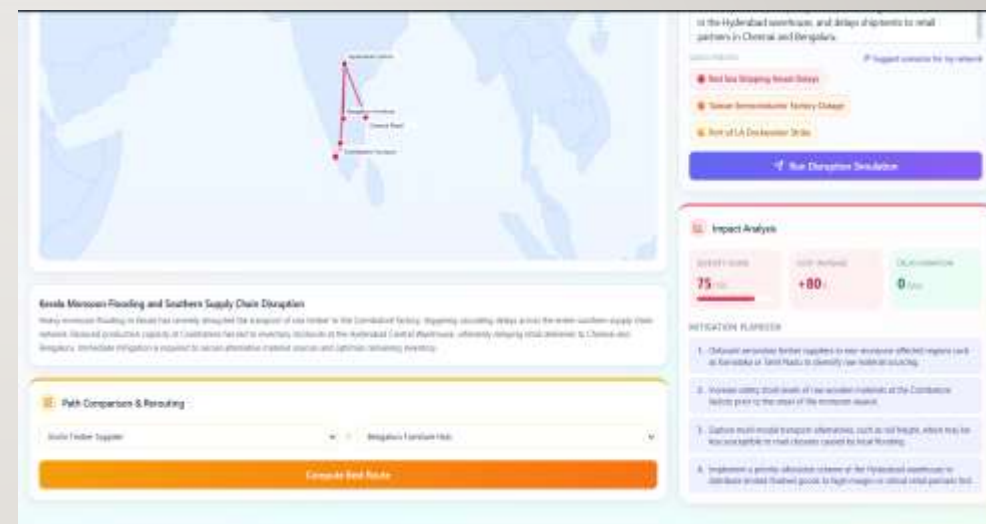
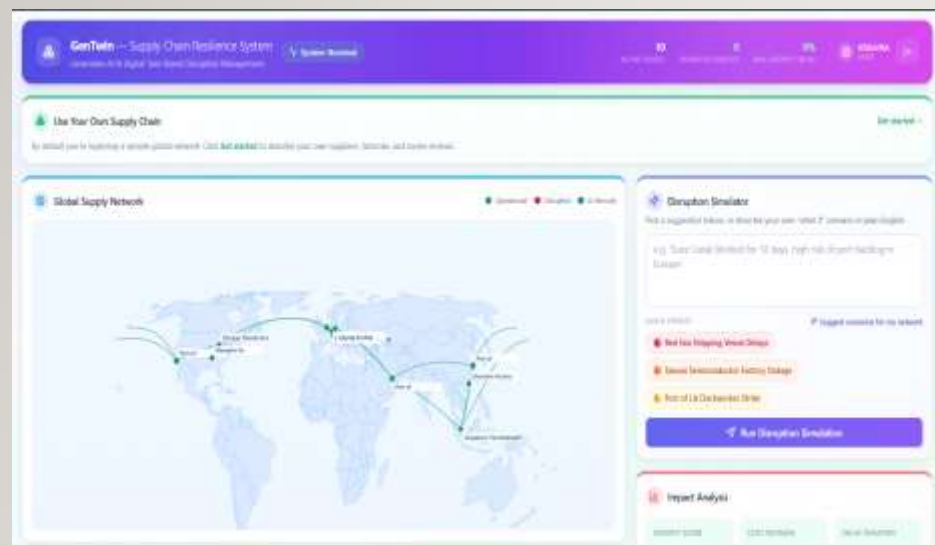
Tool / Technology	Purpose
Python	Backend development and Implementation
Streamlit	Interactive web application development
SQLite	Database management
Pandas & NumPy	Data preprocessing and analysis
NetworkX	Digital Twin and supply chain network modeling
Gemini API	AI-based disruption scenario generation
Plotly / Matplotlib	Data visualization and dashboards



Engineer to Excel



RESULTS



CONCLUSION

- Developed a Generative Digital Twin for simulating supply chain disruptions.
- Used Generative AI to predict and analyze realistic disruption scenarios.
- Evaluated the impact of disruptions on inventory, transportation, and delivery performance.
- Recommended effective resilience strategies for faster recovery and business continuity.
- Improved decision-making, reduced operational risks, and enhanced overall supply chain resilience.
- Provided a scalable framework that can support future AI-driven supply chain management.

REFERENCES

- Benhamou, L., Giard, V., & Lamouri, S. (2026). *Digital Twins in Supply Chain Management: Scope and Methodological Issues*. International Journal of Production Economics.
- Ivanov, D. (2026). *Agentic Digital Twins: Bridging Model-Based and AI-Driven Decision-Making Support*. International Journal of Production Research.
- Abdel Hakim Azzi & Karim Gacem (2026). *A Bibliometric Analysis of Digital Twin Applications in Supply Chain Management: Evidence from Scopus (2015–2025)*.
- Osei, E., et al. (2024). *Supply Chain 4.0 and the Digital Twin Approach: A Framework for Improving Supply Chain Visibility*. Procedia CIRP.
- Le, T. V., & Fan, R. (2023). *Digital Twins for Logistics and Supply Chain Systems: Literature Review, Conceptual Framework, Research Potential, and Practical Challenges*. arXiv.

THANK YOU